



Project Design Document: The Sentinel Hybrid-SOC

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Project Phase: Month 1 - Infrastructure Design

Status: In Progress (Active Development)

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1. Executive Summary

The **Sentinel Hybrid-SOC** is a comprehensive security ecosystem designed to provide unified visibility and automated response across on-premise endpoints and cloud infrastructure. By integrating **Wazuh (SIEM/EDR)**, **AWS (Cloud)**, and **Shuffle (SOAR)**, the project simulates a modern Security Operations Center (SOC) capable of detecting sophisticated multi-stage attacks and containing them without human intervention.

2. Problem Statement

Modern organizations suffer from "Alert Fatigue" and fragmented security visibility. Most entry-level security setups focus solely on local networks, ignoring the risks of cloud credential theft or the need for rapid, automated containment. This project solves these gaps by providing a centralized "single pane of glass" for detection and a "low-latency" response mechanism.

3. Project Objectives (SMART)

- **Visibility:** Centralize logs from Windows 10 endpoints and AWS CloudTrail into a Wazuh Manager.
- **Detection:** Map all simulated attacks to the **MITRE ATT&CK** framework.
- **Automation:** Reduce Mean Time to Respond (MTTR) by automating the isolation of infected hosts using Shuffle SOAR.
- **Reporting:** Generate professional-grade incident reports for every high-severity alert.

4. Technical Architecture

4.1 System Components

- **SIEM/EDR (Wazuh):** Centralized log analysis, file integrity monitoring, and vulnerability assessment.
- **Telemetry (Sysmon):** Deep-level Windows event logging (Process creation, Network connections).
- **Cloud (AWS):** Monitoring for unauthorized API calls and resource manipulation.
- **Orchestration (Shuffle):** Automated workflow engine for alert enrichment and response.

4.2 High-Level Logic (The "Algorithm")

1. **Ingestion:** Windows Agent → Wazuh Manager ← AWS CloudTrail.
2. **Detection:** Alert triggered if rule severity >10.
3. **Enrichment:** Alert forwarded to Shuffle; IP/Hash checked against **VirusTotal**.
4. **Action:** If "Malicious," Shuffle sends a command to Wazuh to **Isolate Host**.

5. Implementation Roadmap

- **Month 1 (Bones):** Virtualization setup, Wazuh Manager installation, and Agent deployment.
- **Month 2 (Nerves):** Simulation of Malware (Endpoint) and Credential Theft (Cloud).
- **Month 3 (Brain):** Workflow automation in Shuffle and final Portfolio documentation.

6. Success Metrics

- **Detection Rate:** 100% of "Atomic Red Team" tests must trigger an alert in Wazuh.
- **Automation Speed:** Containment action (Host Isolation) must occur within **60 seconds** of initial alert.
- **Clarity:** A non-technical manager should be able to understand the "Incident Timeline" report.